Compliance sheet — star-rotframe

Everything a supplier qualification, a legal review or an export check usually asks about star-rotframe 0.1.0, in one document.

1. Identity

Document	STAR-PKG-STAR-ROTFRAME-EN · v1.2 · 2026-10-15
Package	star-rotframe
Version	0.1.0
Capability	star.frames.rotating-z
Maturity	RELEASED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_rotframe

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE and NOTICE files, keep the copyright notices, and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import
hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_rotframe-
0.1.0-py3-none-any.whl
```

FILE	SHA-256
star_rotframe-0.1.0-py3-none-any.whl	0a0beaefee068cd94d5c9cedfadf3f5e8f960cd376dae03d595acdb9dd66acef
star_rotframe.py	a7e8682356b258cd2b7ed91f3f67657d8f0b25ce4afa2ee1c93e8d7b77506cd2
star_rotframe-0.1.0-source.zip	c4c323c177fe1e3df8ca874f6366b103afc221267bf6ad5cf2912257afaf1274

5. What it does, and its limits

Position and velocity between an inertial frame and a frame rotating about z at a given angle and rate, transport term included, in both directions; on 600 states there and back the results agree with NAIF SPICE (rav2xf), SciPy and astropy within 5.2e-16 of |r| and of |v| + |rate| |r|.

A rotation about one fixed axis at a constant rate. For the Earth this leaves out polar motion, precession, nutation and the variation of the length of day: this module is NOT a full ICRS-to-ITRS transformation, and the angle (Earth Rotation Angle or sidereal time) must come from elsewhere (see `star_era` in this repository). Over a day the neglected terms move a ground point by metres to tens of metres. No acceleration (Coriolis and centrifugal terms) is returned. Units are whatever the caller uses, consistently.

- `to_rotating(r, v, angle, rate)` returns r' = Rz(angle) r and v' = Rz(angle) v − w × r', with w = (0, 0, rate): the velocity seen in the rotating frame, transport term included. `angle` is in radians, about +z.
- `to_inertial` is the exact inverse. The default rate is the WGS-84 angular velocity of the Earth, `EARTH_RATE` = 7.292115e-5 rad/s; any other rate, of either sign, can be given.
- Measured on 600 states (orbits in km and in metres, random magnitudes, angles up to 1e8 rad, rates up to 0.1 rad/s), there and back: agreement with NAIF SPICE (`rav2xf`), SciPy and astropy within 5.2e-16 of |r| and of |v| + |rate| |r|; round trip within 3.4e-16.
- Every function returns two tuples of three floats or raises `ValueError`: vectors that are not three finite numbers within 1e15, an angle or a rate that is not finite or is out of range, booleans.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_rotframe_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-rotframe}: Position and velocity between an inertial and a rotating frame},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_rotframe/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts